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1) What are the different protocols you observe at the following layers of protocol stack?

→ After following the given instruction in the document I am able to capture 16700 packets within 6 mins.

i) Application layer: — In 16700 packets, found 497 (3.0%) of Application protocol which include

- HTTP / JSON → Contain 1 packet is less than 0.1%.
- DHCP → Contain 19 packets i.e 0.1%.
- DNS → Contain 2.8% i.e 462 packets — et.
- HTTP → Contain 16 (0.1%) Packets
- SSDP → Contain 0.3% i.e 55 packets.
- Link → local multicast name resolution
→ Contain 20% i.e packets in each IPv4 & IPv6.
- Multicast DNS → 9.5% i.e 430 packets in IPv6 & 2.5% i.e 420 packets in IPv4.
- Dynamic host configuration protocol → Contain 0.1% i.e 19 packets in IPv4 & 0.1% i.e 15 packets in IPv6

2) Transport layer :-

- TCP $\rightarrow$ Contain 38.6% i.e 60393 Packets.
- UDP $\rightarrow$ contain 26.2% i.e 438 Packets in IPv4 & 2.7% i.e 455 Packet in IPv6.

3) Network layer :- internet group management Protocol

- Contain 0.6% i.e 95 packet
- Internet Control message protocol (ICMPv6) $\rightarrow$ Contain 1.6% i.e 248 Packet.
- IPv4 : Contain 65% i.e 10305 Packets
- IPv6 : Contain 4.2% i.e 703 Packets.

2. What is the total amount of data being received for the following two case?

1) Neverssl.com $\rightarrow$ IP Add $\rightarrow$ [34.228.124.45]
Step $\rightarrow$ filter IP address for specific website
used IP add = 34.223.124.40
Step II $\rightarrow$ Statics $\rightarrow$ conversation & then: so found total 200 packet conversion b/w & total amount of data received is if packet which is 32 kB.

2) Cornell.edu : IP Adder $\rightarrow$ [128.253.173.244]
I found total 72 packet of 29kB & total packet received from cornell.edu is 22kB of total packet 36.

Q3. Answer the following When you access the hite neverss.com $\rightarrow$

i) what version of HTTP the server running?
→ only HTTP/1

ii) How many HTTP GET request do you observe?
→ total 6 HTTP GET request from neverssl.com

* HTTP GET request are: -

i) 579 GET /online HTTP/1.1

ii) 580 GET /online HTTP/1.1

iii) 485 GET /javicom.ico HTTP/1.1

iv) 585 GET /online HTTP/1.1

v) 559 GET /online HTTP/1.1

vi) 495 GET /online HTTP/1.1

∴ By using "http request.method == "GET" and IP address = 34.229.124.45

3) Identify TCP three way handshake used to establish the connection with ACK packets & write the sequence no. of each of the three access.

Solve: - Get request

NO → 3295 → GET /online HTTP/1.1

Stream no → 101

<u>Number</u>	<u>information</u>	<u>Sequence no.</u>
3198	SYN	0
3295	SYN.ACK	0
3296	ACK	1

Step which I follow to answer

Step 1 → Get the details of GET HTTP request.

"http.request.method == "GET"

Step 2 → Select any request & check stream index from panel.

Step 3 → To get TCP handshake details filter using this command →

"tcp.stream == 101"

Step 4 → Get the information from into edummm & non-edummm & panel for sequence no.